

CO2
Assessment Tool-1
Short Answers.

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Course Code : ITAD501

Course Name : Computer Vision

1) Sampling:

Low spatial resolution means fewer pixels are used to represent the image. Increase the sampling rate to capture the image more pixels and make small objects clear.

2) Dark and Noisy Image:

Use Histogram equalization to improve the brightness and contrast, and a noise filter to reduce noise. This makes the right image clearer.

3) Quantization:

Quantization converts continuous intensity value into discrete digital levels. It helps a computer to store and process the image digitally.

4) Uneven Lighting:

Uneven lighting can hide defects in some parts of the product. Use contrast enhancement such as CLAHE. It improves the contrast of different regions of the image. Therefore, defects become more visible.

5) Image Segmentation:

Image Segmentation divides an image into meaningful image. It can separate the road, vehicle, and pedestrians. Each object can then be identified separately. It helps a self-driving car understand its surrounding.

6) Edge Detection:

Use Canny edge detection to identify edges in the component. Edges represent sudden changes in image intensity. Crack boundaries can therefore be highlighted. It help the system detect defective.

7) Filtering:

Medical Images may contain unwanted noise. A median filter can be used to reduce the noise. It preserves important edges while smoothing the image. It makes the tumor easier to identify.

8) Object Detection:

Object detection identifies and locates vehicle in a video. The detected vehicles can also be congestion detection. Therefore, it support effective traffic management.

9) Motion Blur:

The blur is caused by movement of the object during image capture. This is called motion blur. It improves the sharpness and visibility of the image.

10) Image Classification:

Image classification categorize product based on their appearance. Products can be classified as good or defective. The system learns visual features and predicts the class. This supports fast and automated quality inspection.